2021 Course Outline
Mathematics Applications – ATAR Year 11
This outline is subject to change

The program you have been issued references all objectives in the School Curriculum and Standards Authority syllabus. The syllabus is available to be downloaded from the School Curriculum and Standards Authority website. (www.scsa.wa.edu.au)

SYLLABUS CONTENT

Unit	Topics	Time Placement
1	1. Consumer Arithmetic	20 hours
	2. Algebra and Matrices	10 hours
	3. Shape and Measurement	20 hours
2	1. Univariate Data Analysis and the Statistical Investigation Process	20 hours
	2. Applications of Trigonometry	10 hours
	3. Linear Equations and their Graphs	20 hours

ASSESSMENT OUTLINE

Assessment Type	Expected Number	Weighting
Major Test	4	40%
Investigations	2	20%
Examinations	2	40%

COURSE OUTLINE

TEXT Sadler Mathematics Applications Unit 1 and 2 (NelsonNet)

WEBSITES School Curriculum and Standards Authority (www.scsa.wa.edu.au)

NOTE Please be aware that assessment times may change throughout the year. Students will be advised of any changes to the assessment schedule.

WEEK	CONTENT AND SYLLABUS REFERENCE	TEXT	ASSESSMENTS
TERM 1 Week 1 Week 2 Week 3	**CONSUMER ARITHMETIC** **Applications of rates and percentages** 1.1.5. Apply percentage increase or decrease in contexts, including determining the impact of inflation on costs and wages over time, calculating percentage mark-ups and discounts, calculating GST, calculating profit or loss in absolute and percentage terms, and calculating simple and compound interest.	*Chapter 2* *UNIT 1* *(Ex 2A – 2C)* *Chapter 3* *UNIT 1* *(Ex 3A – 3C)* *Chapter 4* *UNIT 1* *(Ex 4A – 4B)* *Mathspace*	
Week 4 Week 5	1.1.1. Calculate weekly or monthly wage from an annual salary, wages from an hourly rate, including situations involving overtime and other allowances, and earnings based on commission or piecework. 1.1.2. Calculate payments based on government allowances and pensions. 1.1.3. Prepare a personal budget for a given income taking into account fixed and discretionary spending 1.1.4. Compare prices and values using the unit cost method 1.1.6. Use currency exchange rates to determine the cost in Australian dollars of purchasing a given amount of a foreign currency, or the value of a given amount of foreign currency, when converted to Australian dollars 1.1.7. Calculate the dividend paid on a portfolio of shares given the percentage dividend or dividend paid for each share, and compare share values by calculating a price-to-earnings ratio.	*Chapter 5* *UNIT 1* *(Ex 5A – 5F)* *Mathspace*	**TEST 1: 10%** **Week 5** **(1.1.1 – 1.1.7)**
Week 6 Week 7	**Use of spread sheets** 1.1.8. Use a spreadsheet to display examples of the above computations when multiple or repeated computations are required; for example, preparing a wage-sheet displaying the weekly earnings of workers in a fast food store where hours of employment and hourly rates of pay may differ, preparing a budget, or investigating the potential cost of owning and operating a car over a year. **ALGEBRA AND MATRICES** **Linear and Non-Linear Expressions** 1.2.1. Substitute numerical values into algebraic expressions, and evaluate (with the aid of technology where complicated numerical manipulation is required). 1.2.2. Determine the value of the subjects of a formula, given the values of the other pronumerals in the formula (transposition not required). 1.2.3. Use a spreadsheet or an equivalent technology to construct a table of values from a formula, including tables for formulas with two variable quantities; for example, a table displaying the body mass index (BMI) of people of different weights and heights.	*Chapter 1* *UNIT 1* *(Ex 1A)* *Mathspace*	

Week 8 Week 9	**Matrices and Matrix Arithmetic** 1.2.4. Use matrices for storing and displaying information that can be presented in rows and columns; for example, databases, links in social or road networks. 1.2.5. Recognise different types of matrices (row, column, square, zero, identity) and determine their size. 1.2.6. Perform matrix addition, subtraction, multiplication by a scalar, and matrix multiplication, including determining the power of a matrix using technology with matrix arithmetic capabilities when appropriate. 1.2.7. Use matrices, including matrix products and powers of matrices, to model and solve problems; for example, costing or pricing problems, squaring a matrix to determine the number of ways pairs of people in a communication network can communicate with each other via a third person.	*Chapter 6* *UNIT 1* *(Ex 6A – 6C)* *Mathspace*	**INV 1: 6%** **OUT: Week 7** **IN: Week 9** **(1.1.8)**
TERM 2 Week 1 Week 2 Week 3	**SHAPE AND MEASUREMENT** **Pythagoras' Theorem** 1.3.1. Use Pythagoras' theorem to solve practical problems in two dimensions and for simple applications in three dimensions. **Mensuration** 1.3.2. Solve practical problems requiring the calculation of perimeters and areas of circles, sectors of circles, triangles rectangles, parallelograms and composites. 1.3.3. Calculate the volumes of standard three-dimensional objects, such as spheres, rectangular prisms, cylinders, cones, pyramids and composites in practical situations, for example, the volume of water contained in a swimming pool. 1.3.4. Calculate the surface areas of standard three-dimensional objects, such as spheres, rectangular prisms, cylinders, cones, pyramids and composites in practical situations; for example, the surface area of a cylindrical food container.	*Chapter 7* *UNIT 1* *(Ex 7A – 7B)* *Chapter 8* *UNIT 1* *(Ex 8A – 8B)* *Chapter 9* *UNIT 1* *(Ex 9A – 9D)* *Chapter 10* *UNIT 1* *(Ex 10A)* *Mathspace*	**TEST 2: 10%** **Week 3** **(1.2.1 – 1.3.4)**
Week 4 Week 5	**Similar Figures and Scale Factors** 1.3.5. Review the conditions for similarity of for two-dimensional figures, including similar triangles. 1.3.6. Use the scale factor for two similar figures to solve linear scaling problems. 1.3.7. Obtain measurements from scale drawings, such as maps or building plans, to solve problems. 1.3.8. Obtain a scale factor and use it to solve scaling problems involving the calculation of the areas of similar figures and surface areas and volumes of similar solids.	*Chapter 10* *UNIT 1* *(Ex 10A – 10D)* *Mathspace*	
Week 6 Week 7	**Revision for Exams**		
Week 8 Week 9	**SEMESTER ONE EXAMINATIONS | UNIT 1**		**Semester One Examination (ALL of Unit 1) 18%**

Week 10 Week 11	**UNIVARIATE DATA ANALYSIS AND THE STATISTICAL INVESTIGATION PROCESS** **The Statistical Investigation Process** 2.1.1. Review the statistical investigation process; identifying a problem and posing a statistical question, collecting or obtaining data, analysing the data, interpreting and communicating the results. **Making Sense of Data Relating to a Single Statistical Variable** 2.1.2. Classify a categorical variable as ordinal, such as income level (high, medium, low) or nominal, such as place of birth (Australia, overseas) and use tables and bar charts to organise and display data. 2.1.3. Classify a numerical variable as discrete, such as the number of rooms in a house, or continuous, such as the temperature in degrees Celsius. 2.1.4. With the aid of an appropriate graphical display (chosen from dot pot, stem plot, bar chart or histogram), describe the distribution of a numerical dataset in terms of modality (uni or multimodal), shape (symmetric versus positively or negatively skewed), location and spread and outliers, and interpret this information in the context of the data. 2.1.5. Determine the mean and standard deviation of a dataset using technology and use these statistics as measures of location and spread of data distribution, being aware of their limitations.	*Chapter 1* *UNIT 2* *(Ex 1A – 1C)* *Chapter 2* *UNIT 2* *(Ex 2A – 2C)* *Chapter 3* *UNIT 2* *(Ex 3A – 3C)*	
<u>TERM 3</u> Week 1 Week 2 Week 3	**Making Sense of Data Relating to a Single Statistical Variable** 2.1.4. With the aid of an appropriate graphical display (chosen from dot pot, stem plot, bar chart or histogram), describe the distribution of a numerical dataset in terms of modality (uni or multimodal), shape (symmetric versus positively or negatively skewed), location and spread and outliers, and interpret this information in the context of the data. 2.1.5. Determine the mean and standard deviation of a dataset using technology and use these statistics as measures of location and spread of data distribution, being aware of their limitations. 2.1.6. Use the numbers of deviations from the mean (standard scores) to describe deviations from the mean in normally distributed data sets. 2.1.7. Calculate quantiles for normally distributed data with known mean and standard deviation in practical situations. 2.1.8. Use the 68%, 95% and 99.7% rule for data one, two and three standard deviations from the mean in practical situations. 2.1.9. Calculate the probabilities for normal distributions with known mean and standard deviation in practical situations. **Comparing Data for a Numerical Variable Across Two or More Groups** 2.1.10. Construct and use parallel box plots (including the use of the 'Q1 – 1.5 2.1.11. Compare groups on a single numerical variable using medians, means, IQRS, ranges or standard deviations, and as appropriate; interpret the differences observed in the context of the data and report the findings in a systematic and concise manner. 2.1.12. Implement the statistical investigation process to answer questions that involve comparing the data for a numerical variable across two or more groups; for example, are Year 11 students the fittest in the school?	*Chapter 4* *UNIT 2* *(Ex 4A – 4B)* *Chapter 13* *(13A – 13C)*	**INV 2: 10%** **OUT: Week 1** **IN: Week 2** **(The Statistical Investigation Process)**

Week 4 Week 5	**APPLICATIONS OF TRIGONOMETRY** 2.2.1. Use trigonometric ratios to determine the length of an unknown side, or the size of an unknown angle in a right-angled triangle. 2.2.2. Determine the area of a triangle, given two sides and an included angle by using the rule $Area = \frac{1}{2}ab\sin C$, or given three sides by using Heron's Rule, and solve related practical problems. 2.2.3. Solve problems involving non-right angled triangles using the Sine Rule (acute triangles only when determining the size of an angle) and the Cosine Rule. 2.2.4. Solve practical problems involving right-angled and non-right angled triangles, including problems involving angles of elevation and depression and the use of bearings in navigation.	*Chapter 10 UNIT 2 (Ex 10A – 10B)* *Chapter 11 UNIT 2 (Ex 11A – 11C)* *Mathspace*	**TEST 3: 10% Week 4 (2.1.1 – 2.1.12)**
Week 6 Week 7 Week 8 Week 9	**LINEAR EQUATIONS AND THEIR GRAPHS** **Linear Equations** 2.3.1. Identify and solve linear equations (with the aid of technology where complicated manipulations are required). 2.3.2. Develop a linear formula from a word description and solve the resulting equation. **Straight-line graphs and their applications** 2.3.3. Construct straight-line graphs both with and without the aid of technology. 2.3.4. Determine the slope and intercepts of a straight-line graph from both its equation and its plot. 2.3.5. Construct and analyse a straight-line graph to model a given linear relationship; for example, modelling the cost of filling a fuel tank of a car against the number of litres of petrol required. 2.3.6. Interpret, in context, the slope and intercept of a straight-line graph used to model and analyse a practical situation. **Piece-wise linear graphs and step graphs** 2.3.9. Sketch piece-wise linear graphs and step graphs, using technology when appropriate. 2.3.10. Interpret piece-wise linear and step graphs used to model practical situations; for example, the tax paid as income increases, the change in the level of water in a tank over time when water is drawn off at different intervals and for different periods of time, the charging scheme for sending parcels of different weight through the post.	*Chapter 6 UNIT 2 (Ex 6A)* *Chapter 7 UNIT 2 (Ex 7A – 7C)* *Chapter 8 UNIT 2 (Ex 8A – 8D)* *Chapter 9 UNIT 2 (Ex 9A)* *Mathspace*	**TEST 4: 10% Week 8 (2.2.1 – 2.3.10)**
Week 10	**Simultaneous linear equations and their applications** 2.3.7. Solve a pair of simultaneous linear equations graphically or algebraically, using technology when appropriate. 2.3.8. Solve practical problems that involve determining the point of intersection of two straight-line graphs; for example, determining the break-even point where cost and revenue are represented by linear equations.	*Chapter 12 UNIT 2 (Ex 12A)* *Mathspace*	

<u>Term 4</u> **Week 1** **Week 2**	**Review / Exam Preparation**		
Week 3 **Week 4**	**Semester Two Examination| Unit 1 and 2**		**Semester Two Examination (ALL of Unit 1 and 2) 22%**

Appendix 1 – Grade descriptions Year 11

Grade	Description
A	**Identifies and organises relevant information** Identifies and organises relevant information that is complex; for example, developing a linear formula from a word description, interpreting word problems to determine the appropriate pension allowance or dividend amount for a share portfolio. Uses scale to calculate appropriate areas and volumes of similar shapes and solids, drawing geometrical diagrams from descriptive passages.
	Chooses effective models and methods and carries through the methods correctly Solves extended unstructured problems; for example, adding the correct information to a given geometry diagram or recognising gradient and intercept as part of linear graph used to model a practical situation. Carries extended responses through; for example, developing a diagram and using the result to solve related problems or using extended tables or a spreadsheet to find patterns both in the rows and columns. Determines the effects of changed conditions; for example, recognising the effects of change of gradient or intercept on a linear graph.
	Follows mathematical conventions and attends to accuracy Defines and uses variables appropriately in solving algebraic problems and in representing and analysing statistical data. Decides at which point to round in an extended response and determines the degree of accuracy based upon the context or units used when solving problems requiring trigonometry or measurement calculations.
	Links mathematical results to data and contexts to reach reasonable conclusions Generalises mathematical structures and applies appropriate processes, such as, the statistical investigative process, to answer questions in context. Makes appropriate use of the scale on maps. Recognises specified conditions and attends to units in extended responses.
	Communicates mathematical reasoning, results and conclusions Shows clear steps in reasoning; for example, when solving a problem using matrices. Justifies conclusions with a statement which links to results. Translates solution into language of the original problem.

Grade	Description
B	**Identifies and organises relevant information** Identifies and organises relevant information that is concentrated or scattered; for example, accurately labelling 2-D or simple 3-D diagrams with part information included; completing tables of values to look for a pattern or preparing a wage-sheet; and identifying the correct information from a given geometry diagram.
	Chooses effective models and methods and carries the methods through correctly Recognises the correct function model and reads the correct values from a graph. Carries through and solves multi-step linear equations; extracts correct information from a network diagram; extracts correct information from an extended table of values; applies appropriate matrix algebra to solve costing/pricing problems or communication network problems. Generalises obvious mathematical structures; for example, using interpolation/extrapolation appropriately in graphing, finding obvious number patterns in tables of values.
	Follows mathematical conventions and attends to accuracy. Applies conventions for diagrams and graphs including accurately labelling angles and sides in an extended 2D diagram. Rounds to specified accuracies; for example, when calculating price-to-earnings share ratios or performing currency conversions. Checks results and makes adjustments where necessary.
	Links mathematical results to data and contexts to reach reasonable conclusions Recognises specified conditions and attends to units in extended problems. Links and processes more than one piece of information which may be scattered, such as, comparing data across two or more groups and drawing reasonable conclusions.
	Communicates mathematical reasoning, results and conclusions Shows the main steps in reasoning; for example, when solving trigonometric ratio problems from set diagrams. Uses routine methods in labelling networks to show results. Shows main steps in finding a solution to a problem; for example, using Pythagoras' theorem to solve a 2D practical problem.

Grade	Description
C	**Identifies and organises relevant information** Identifies and organises relevant information that is grouped together or is relatively narrow in scope; for example, makes direct substitution of values into linear equations/formulas. Selects the correct sides when using trigonometric ratios on a diagram that is supplied. Identifies the key parameters of a normally distributed data set.

	Identifies the need to determine the point of intersection of two straight line graphs to solve a cost/revenue break-even point problem.
	Chooses effective models and methods and carries the methods through correctly Answers structured questions that require short responses; for example, solves two-step linear equations. Completes missing values in a table of a given linear function or step-graph. Applies mathematics in practised ways; for example, chooses the correct trigonometric ratio for a supplied diagram or to solve a right-triangle problem. Calculates specific cases of generalisations; for example, substitutes values into a given formulas, such as, trigonometric ratios, and evaluates the unknown angle or side. Carries a single thread of reasoning through; for example, applies the area formula, $Area\ \Delta ABC = \frac{1}{2} \times 7 \times 6 \times sin23^o$. Uses a calculator appropriately for calculations.
	Follows mathematical conventions and attends to accuracy Applies the rule of Order of Operations to equations; for example, Pythagoras' theorem. Applies conventions for diagrams and labels sides and angles in geometry. Sets up graphs neatly and accurately and labels histograms appropriately with the scales marked. Rounds to suit contexts; for example, shows dollar calculations to two decimal places and rounds distance to a sensible level with measurement questions. Accurately plots and labels given points when graphing linear functions.
	Links mathematical results to data and contexts to reach reasonable conclusions Recognises specified conditions and attends to units in short responses; for example, expresses the answer using the units defined in the question or links trigonometric ratios to distances on a diagram. Links data to the respective numbers in a matrix.
	Communicates mathematical reasoning, results and conclusions Shows working, including intermediate steps and/or expressions entered into a calculator, when setting out short responses. Uses the 'left hand side = right hand side' convention properly in short responses. Makes comparisons between groups of data, communicating observations in context.

D	**Identifies and organises relevant information** Identifies and organises relevant information that is grouped together and narrow in scope; for example, plotting points on a Cartesian plane or making single value substitutions in a short response. Identifies the appropriate parameters to describe a numerical dataset.
	Chooses effective models and methods and carries the methods through correctly Answers structured questions that are familiar or require short responses; for example, drawing single geometric figures by connecting points on a Cartesian plane. Substitutes into appropriate mensuration formulas to evaluate volume/area of standard shapes/objects. Makes single-step, common sense connections; for example, recognises the use of the constant in a linear equation within a context. Uses a calculator to complete practised processes, such as, determining the mean or standard deviation of a data set.
	Follows mathematical conventions and attends to accuracy Plots graphs with a poor degree of accuracy, or labels diagrams and given points with little detail. Rounds to suit contexts in short answer questions, but only when prompted.
	Links mathematical results to data and contexts to reach reasonable conclusions Recognises specified conditions and attends to units in short responses only in familiar and practised questions. Links calculation results to a context in obvious ways; for example, when linking the calculation of a weekly wage to an hourly rate.
	Communicates mathematical reasoning, results and conclusions Shows working but only in familiar and practised contexts; for example, when calculating the area of a triangle using a provided formula and obvious dimensions.

E	Does not meet the requirements of a D grade and/or has completed insufficient assessment tasks to be assigned a higher grade.